Formalizing Bundle Completion

A probabilistic–deterministic view of BundleFuse

Study note

Abstract

This note answers one question precisely: *is* BundleFuse *a stochastic model or a deterministic function?* The short answer is **both, at different layers**: it is a *deterministic, closed-form predictor* whose finitely many parameters are *estimated from a stochastic data-generating process.* We build the formalism from the ground up—objects, the probability space, the deterministic scoring map, the one probability model it posits (and why that model is rejected), and finally learning-as-estimation—so that each layer's role is unambiguous. A running 4-item toy example is carried throughout, and the Normal–Normal shrinkage identity is derived in full.

Contents

1 Objects and notation **1**

2 The task, and the question this note settles **2**

3 The stochastic layer (S): data and objective **2**

4 The deterministic layer (D): the scoring map **3**

5 The empirical-Bayes model: posited, then rejected **4**

6 Learning as estimation **5**

7 Synthesis **6**

1 Objects and notation

We work with a finite item **catalogue**

$$\mathcal{I} = \{1, 2, \ldots, n\}.$$

A **bundle** is a subset of items that belong together; we only ever consider bundles of size at least two, so the space of bundles is the finite family

$$\mathcal{S} = \{\, b \subseteq \mathcal{I} : |b| \geq 2 \,\}.$$

Training data is a multiset of bundles $\mathcal{B} = \{b_1, \ldots, b_m\}$, which we encode once and for all as a binary **incidence matrix**

$$B \in \{0,1\}^{m \times n}, \qquad B_{ji} = \mathbb{1}[\, i \in b_j \,].$$

Two derived quantities recur. The **item-frequency vector**

$$f = B^\top \mathbf{1}_m \in \mathbb{Z}^n_{\geq 0}, \qquad f_i = \sum_{j=1}^{m} B_{ji}$$

counts how many training bundles contain item i (an item with $f_i = 0$ is called *cold*). For any set $A \subseteq \mathcal{I}$, its **indicator vector** is $x_A \in \{0,1\}^n$ with $(x_A)_i = \mathbb{1}[i \in A]$.

Example 1 (toy catalogue, used throughout). Let $n = 4$ and take three training bundles

$$b_1 = \{1,2,3\}, \quad b_2 = \{1,2,4\}, \quad b_3 = \{2,3\}.$$

Then

$$B = \begin{pmatrix} 1 & 1 & 1 & 0 \\ 1 & 1 & 0 & 1 \\ 0 & 1 & 1 & 0 \end{pmatrix}, \qquad f = B^\top \mathbf{1}_3 = \begin{pmatrix} 2 \\ 3 \\ 2 \\ 1 \end{pmatrix}.$$

Item 2 is the most frequent; every item is warm ($f_i > 0$) here.

2 The task, and the question this note settles

The completion task. At test time we observe a *partial* bundle S (a set of "seed" items) and must rank the remaining catalogue $\mathcal{I} \setminus S$ so that the held-out members of the true bundle appear near the top. Formally, a **ranker** is a scoring function

$$s : \mathcal{S} \times \mathcal{I} \to \mathbb{R}, \qquad (S, c) \mapsto s(c \mid S),$$

which orders the candidates $c \in \mathcal{I} \setminus S$ by decreasing $s(c \mid S)$.

The question. Is such a ranker a *stochastic model* (with a sample space and random variables), or a *deterministic function*? We will see that the honest answer requires separating two layers:

(S) a **stochastic layer** — the data-generating process and the population objective (§3);

(D) a **deterministic layer** — the closed-form scoring map used at inference (§4);

plus one subtlety: the only explicit probability *model* inside the method (empirical Bayes, §5) is posited and then *rejected by the data*, so it operates as a deterministic rule.

3 The stochastic layer (S): data and objective

Probability space and the random bundle. Fix a probability space $(\Omega, \mathcal{F}, \mathbb{P})$. A bundle is a **random variable**

$$\mathbf{b} : \Omega \to \mathcal{S}, \qquad \text{with law} \quad \pi(b) = \mathbb{P}(\mathbf{b} = b),$$

i.e. π is a probability mass function on the finite set $\mathcal{S}$. Intuitively π encodes "which sets of items genuinely go together" in the domain (outfits, playlists, baskets).

Definition 1 (sample space of the learning problem)**.** *The training set is an i.i.d. sample* $\mathbf{b}_1, \ldots, \mathbf{b}_m \overset{iid}{\sim} \pi$*, and a test bundle is an independent draw* $\mathbf{b}^\star \sim \pi$*. All derived objects—the incidence matrix* B*, the frequencies* f*—are measurable functions of this sample and are therefore* themselves random.

The masking random variable. The partial query is produced by hiding part of the test bundle. Let $\mathbf{H}$ be a random held-out subset:

$$\mathbf{H} \mid \mathbf{b}^\star \;\sim\; \text{Uniform over the } \big\lfloor \rho\,|\mathbf{b}^\star| \big\rfloor\text{-subsets of } \mathbf{b}^\star, \qquad \rho = \tfrac{1}{2},$$

and the observed seeds are $S = \mathbf{b}^\star \setminus \mathbf{H}$. Thus the **random variables** of the problem are the bundles $\{\mathbf{b}_j\}$, the test bundle $\mathbf{b}^\star$, and the mask $\mathbf{H}$; the **sample space** is the product Ω carrying all of them.

The population objective. Quality is an *expectation* over this process. With $\mathrm{top}_K(s; S) = \arg\max^{(K)}_{c \notin S} s(c \mid S)$ the K highest-scoring candidates, the **population risk** (expected Recall@K) is

$$\boxed{\mathcal{R}_K(s) = \mathbb{E}_{(S,\mathbf{H})}\left[\frac{\big|\,\mathrm{top}_K(s; S) \cap \mathbf{H}\big|}{|\mathbf{H}|}\right].} \tag{1}$$

This is the only place probability enters the goal. Learning means choosing s to maximise (1); since π is unknown we maximise an empirical average over a finite held-out sample (§6).

4 The deterministic layer (D): the scoring map

Given the (random) training sample, BundleFuse fixes a parametric family s_θ and the map that turns (B, S) into a score is a **deterministic composition of linear-algebraic operators**. We go through it in order; everything below is a fixed function of x_S.

(a) Collaborative signal — linear in x_S. From the co-occurrence matrix $C = B^\top B - \mathrm{diag}(B^\top B)$ we form the degree-normalised pairwise transition operator

$$W_{\text{pair}} = B^\top \,\mathrm{diag}\big((|b|+1)^{-\gamma}\big)\, B \,\mathrm{diag}\big((f+\alpha_0)^{-\beta}\big), \qquad \mathrm{diag}(W_{\text{pair}}) = 0,$$

and the raw collaborative score is the **linear map**

$$y_{\text{pair}}(\cdot \mid S) = W_{\text{pair}}\, x_S \in \mathbb{R}^n. \tag{2}$$

A second fixed operator $\mathcal{P}$ (a zero-diagonal bipartite walk) gives two-hop smoothing $y_{\text{diff2}} = \mathcal{P}\, y_{\text{pair}}$.

(b) Set-conditioned prior — linear in x_S. Taxonomic and content operators (category PMI P_{cat}, signed slot-exclusion, category-disentangled content $\tilde{X}$, graph bridging) are assembled into a fixed operator Φ so that

$$\mu_0(\cdot \mid S) = \Phi\, x_S \in \mathbb{R}^n. \tag{3}$$

Crucially Φ needs *no* collaborative exposure of the candidate, so μ_0 is informative even for cold items ($f_c = 0$), where every collaborative channel in (2) is identically zero.

(c) Standardise and fuse — linear in the weights. Each signal $k \in \{\text{pair}, \text{diff2}, \text{cat}, \dots\}$ is column-standardised, $Z_k = \mathrm{std}(y_k)$, and fused with weights $w = (w_k)$:

$$Y(\cdot \mid S) = \sum_k w_k\, Z_k \in \mathbb{R}^n. \tag{4}$$

(d) Frequency-confidence routing — diagonal reweighting. With confidence weights $u_c = \dfrac{\kappa}{f_c + \kappa}$ and $U = \mathrm{diag}(u)$,

$$\boxed{\hat{\theta}(\cdot \mid S) = (I - U)\, Y + U\, \mu_0} \qquad \Longleftrightarrow \qquad \hat{\theta}_c = (1 - u_c)\, Y_c + u_c\, \mu_{0,c}. \tag{5}$$

For a cold item $f_c = 0 \Rightarrow u_c = 1$, so $\hat{\theta}_c = \mu_{0,c}$: cold candidates are ranked purely by the prior. For a very frequent item $u_c \to 0$, so $\hat{\theta}_c \to Y_c$: warm candidates trust the collaborative evidence.

(e) Rank. Finally $\mathrm{top}_K(s_\theta; S) = \arg\max^{(K)}_{c \notin S} \hat{\theta}_c$.

Proposition 1 (the predictor is a deterministic map)**.** *For fixed parameters $\theta = (w, \kappa, \dots)$, steps (a)–(e) define a deterministic, measurable function*

$$s_\theta(\cdot \mid S) = g(B, S; \theta),$$

affine–linear in x_S up to the final arg max*. No randomness is sampled at inference time.*

Remark 1 (batch invariance: an even stronger determinism)*.* Because the standardisation scale of the prior is *frozen on the validation split* rather than recomputed per test batch, $s_\theta(c \mid S)$ depends only on S—never on which other queries are scored alongside it. There is thus not even implementation-level nondeterminism (verified to 0.000000 cross-query drift).

Example 2 (one deterministic score, continuing Example 1)*.* Take the raw co-occurrence signal C (set $\gamma = \beta = 0$ for illustration, so $W_{\text{pair}} = C$). From Example 1,

$$B^\top B = \begin{pmatrix} 2 & 2 & 1 & 1 \\ 2 & 3 & 2 & 1 \\ 1 & 2 & 2 & 0 \\ 1 & 1 & 0 & 1 \end{pmatrix}, \qquad C = \begin{pmatrix} 0 & 2 & 1 & 1 \\ 2 & 0 & 2 & 1 \\ 1 & 2 & 0 & 0 \\ 1 & 1 & 0 & 0 \end{pmatrix}.$$

Query the partial bundle $S = \{1\}$, i.e. $x_S = (1, 0, 0, 0)^\top$. The score vector is the deterministic matrix–vector product

$$y = C\, x_S = (\text{column 1 of } C) = \begin{pmatrix} 0 \\ 2 \\ 1 \\ 1 \end{pmatrix}.$$

Excluding the seed, the ranking is 2 (score 2) $\succ$ 3 $\sim$ 4 (score 1): item 2, which co-occurs with item 1 in both b_1 and b_2, is predicted first. No probabilities were drawn—just one linear map.

5 The empirical-Bayes model: posited, then rejected

The *form* of the routing rule (5) is not ad hoc: it is the posterior mean of a conjugate Normal–Normal model. This is the one genuinely probabilistic *model* inside the method, so we derive it and then show the data rejects it.

The model. For a fixed query, treat the (unknown) true compatibility $\theta_{S,c}$ as drawn from a prior centred at the metadata prior μ_0, and the observed collaborative score Y as a noisy measurement whose precision grows with exposure f_c:

$$\theta_{S,c} \mid \mu_0 \;\sim\; \mathcal{N}(\mu_0, \sigma_0^2), \qquad Y \mid \theta_{S,c}, f_c \;\sim\; \mathcal{N}\Big(\theta_{S,c},\; \tfrac{\sigma_\epsilon^2}{f_c}\Big). \tag{6}$$

Proposition 2 (Normal–Normal posterior mean ⇒ Eq. (5))**.** *Under* (6)*, the posterior mean of* $\theta_{S,c}$ *given* Y *is the convex combination*

$$\mathbb{E}[\theta_{S,c} \mid Y] = (1 - u_c)\, Y + u_c\, \mu_0, \qquad u_c = \frac{\kappa}{f_c + \kappa}, \qquad \kappa = \frac{\sigma_\epsilon^2}{\sigma_0^2}.$$

Proof. Gaussian priors are conjugate to Gaussian likelihoods, so the posterior is Gaussian and we only need its mean, which equals the *precision-weighted average* of the two information sources. The prior contributes precision $1/\sigma_0^2$ about μ_0; the likelihood contributes precision f_c/σ_ϵ^2 about Y (the $1/f_c$ variance inflates precision by f_c). Hence

$$\mathbb{E}[\theta \mid Y] = \frac{\dfrac{1}{\sigma_0^2}\,\mu_0 + \dfrac{f_c}{\sigma_\epsilon^2}\,Y}{\dfrac{1}{\sigma_0^2} + \dfrac{f_c}{\sigma_\epsilon^2}}.$$

Multiply numerator and denominator by $\sigma_0^2\sigma_\epsilon^2$:

$$= \frac{\sigma_\epsilon^2\,\mu_0 + f_c\,\sigma_0^2\,Y}{\sigma_\epsilon^2 + f_c\,\sigma_0^2}.$$

Now divide through by σ_0^2 and set $\kappa = \sigma_\epsilon^2/\sigma_0^2$:

$$= \frac{\kappa\,\mu_0 + f_c\,Y}{\kappa + f_c} = \underbrace{\frac{f_c}{f_c + \kappa}}_{1-u_c}\, Y + \underbrace{\frac{\kappa}{f_c + \kappa}}_{u_c}\, \mu_0. \qquad \square$$

So $\kappa = \sigma_\epsilon^2/\sigma_0^2$ is a *ratio of variances*: large measurement noise σ_ϵ^2 (trust the prior more) or small prior spread σ_0^2 (prior is confident) both increase shrinkage.

The testable consequence, and its failure. Model (6) implies the homoscedastic moment condition

$$v_c := \mathbb{E}_S\big[(Y - \mu_0)^2\big] = \sigma_0^2 + \sigma_\epsilon^2 \cdot \frac{1}{f_c},$$

so regressing v_c on $1/f_c$ over warm items should yield slope $\sigma_\epsilon^2 > 0$. **Empirically the OLS slope is negative on 7 of 8 datasets** (zero-inflated collaborative scores: rare items have near-zero dispersion, hubs have large dispersion—the opposite of the model). The model is therefore *rejected*: we do *not* estimate κ from (6). Instead κ is treated as a single scalar hyperparameter selected deterministically on validation (§6). *The probabilistic layer thus collapses to a deterministic frequency-confidence interpolation whose form it merely motivates.*

6 Learning as estimation

Parameters are *fit*, not sampled, in two deterministic stages applied to random data.

Stage 1 — signal construction. The map $B \mapsto (W_{\text{pair}}, \mathcal{P}, \Phi, \dots)$ is a deterministic function of the training sample.

Stage 2 — empirical risk maximisation on validation. Let $\mathcal{V} = \{(S_i, H_i)\}_{i=1}^{v}$ be the held-out validation queries. We choose

$$\hat{\theta} = \arg\max_{\theta \in \Theta} \widehat{\mathcal{R}_K}^{\mathcal{V}}(s_\theta), \qquad \widehat{\mathcal{R}_K}^{\mathcal{V}}(s_\theta) = \frac{1}{v} \sum_{i=1}^{v} \frac{\left|\operatorname{top}_K(s_\theta; S_i) \cap H_i\right|}{|H_i|}, \tag{7}$$

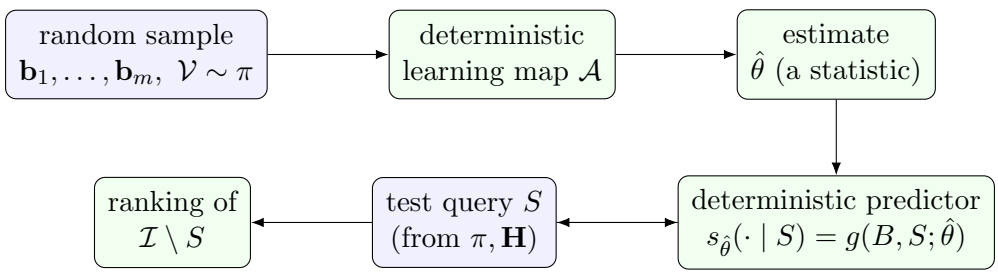
over a small finite grid Θ (greedy coordinate ascent on the weights w, the higher-order mixing, and κ). Here $\widehat{\mathcal{R}_K}^{\mathcal{V}}$ is the *empirical estimate* of the population risk (1).

Remark 2 (the surrogate actually optimised, and the role of K). In code the selection objective is not the raw Recall@K of (7) but validation *NDCG@20*: a monotone, position-discounted refinement of the same top-K functional that is far smoother over the discrete grid Θ (Recall@K is piecewise constant, so many grid points tie; NDCG breaks those ties toward sharper rankings). This is a deliberate surrogate choice, not a different goal: the target remains top-K retrieval at the *deployment operating point* (a short recommendation slate). Two facts make it principled rather than self-referential. **(i)** The estimate is empirically insensitive to the cutoff: selecting at NDCG@K for $K \in \{10, 20, 50, 100\}$ yields the same test Recall@20 (a flat plateau), and selecting *directly* on Recall@20 is no better. **(ii)** Genuinely cutoff-free objectives are *worse*: a softmax negative-log-likelihood (a proper scoring rule for the whole distribution) runs ∼7–31% below, and mean reciprocal rank is regime-dependent and can collapse on sparse data, because neither concentrates on the operating point. Thus K is a property of the deployment target, not a tuned knob, and the exact value is immaterial within a wide band (see the selection-sensitivity study, `results/selection_sensitivity.md`).

Where the output is random. This is the *only* sense in which BundleFuse's output is stochastic: $\hat{\theta} = \hat{\theta}(\mathcal{B}, \mathcal{V})$ is a **statistic**—a measurable function of the random sample—hence a random variable. The relevant questions are then estimation-theoretic:

- **Generalisation gap** $\left|\mathcal{R}_K(s_{\hat{\theta}}) - \widehat{\mathcal{R}_K}^{\mathcal{V}}(s_{\hat{\theta}})\right|$, shown empirically to be ≈ 0 (selection does not overfit);
- **Data efficiency**: how $\hat{\theta}(\mathcal{B})$ degrades as $m = |\mathcal{B}|$ shrinks—measured to be *flat* down to roughly half the data on POG.

7 Synthesis



Blue = stochastic (data process); **green** = deterministic (maps).

The verdict.

- **Randomness lives only in the data**: the law π, the mask $\mathbf{H}$, and the i.i.d. sampling. It defines the population objective $\mathcal{R}_K$ (1) we estimate and makes $\hat{\theta}$ a random estimator.
- **The learning map $\mathcal{A}$ and the predictor g are deterministic.** Given the data, the ranking of $\mathcal{I} \setminus S$ is a fixed (affine-linear-plus-arg max) function of S; batch invariance removes even implementation nondeterminism.
- **The one probability model inside the method (Normal–Normal EB) is posited and rejected**, so in practice $\hat{\theta}$ is an ERM point estimate and $s_{\hat{\theta}}$ a deterministic closed-form map.

In one sentence. BundleFuse is a deterministic, closed-form (predominantly linear-operator) predictor whose finitely many parameters are an empirical-risk estimate from an i.i.d. bundle-generating process; it is "stochastic" only as a statistical *estimator* targeting a population recall functional, not as a sampler.